Proposal: Active Demonstration Querying and Misspecification Detection for MORL

Mason duBoef

Problem

Multi-objective reinforcement learning (MORL) produces a Pareto frontier of policies representing different trade-offs among objectives. The DWPI algorithm [1] infers a user's preference weights from demonstrations by mapping return vectors to weight vectors via a learned model. However, DWPI has two limitations: **(1)** it is susceptible to spurious correlations and **(2)** it assumes all objectives are known and linearly combined, so when the objective space is misspecified, DWPI simply assigns the frontier policy with the closest return profile. Prior work addresses misspecified objectives in single-objective robot learning [2] and hidden context in LLM preference learning [3], but not in demonstration-based MORL. To minimize spurious correlations, I propose querying demonstrations from discriminative starting states. To identify misspecification, I propose residual and max likelihood-based detection. This could be useful in many settings like self-driving cars. Driving in simulation can serve as a natural avenue for inferring driving preferences. However, we want to avoid spurious correlations and recognize misspecification. For example, we might recognize speed and safety as objectives but miss drivers minimizing carsickness.

Methodology

Active Demonstration Querying: After an initial DWPI inference narrows candidate preferences, we identify states where top-k candidate policies diverge most (Jensen-Shannon divergence over actions) and query the user for a demonstration from that state. This is similar to uncertainty sampling in active learning [5]. By querying at maximally discriminative states, we resolve ambiguity from correlated returns with fewer demonstrations.

Misspecification Detection. I will compute the residual between demonstrated returns and the assigned policy's expected returns, then evaluate demonstration likelihood under a Boltzmann rationality model across all frontier policies. The residual and maximum likelihood will be combined into a misspecification score. Ideally, I could distinguish between different types of misspecification (structural, function & observational). I may be able to adapt relevance-gating from Bobu et al. [2] to enable this.

Hypotheses

1: ADQ requires fewer demonstrations than passive querying for equivalent inference accuracy, especially under correlated objectives.

2: Our severity score correlates (Spearman) with true misspecification severity across three types of misspecification, outperforming baseline severity estimator.

Experiments

I'll use the Convex Deep Sea Treasure (2 objectives) and Traffic (5 objectives) benchmarks used in the DWPI paper [1]. Synthetic stochastically-optimal demonstrators with known ground-truth preferences will generate demonstrations.

(1) Compare dynamic demo querying to normal static demonstrations as a basis for inferring preferences. I will evaluate how far the inferred preferences are from the ground-truth using mean squared error.

(2) Induce three misspecification types: (a) structural: hide one of Traffic's 5 objectives; (b) functional: demonstrators optimize a nonlinear objective combination; (c) observational: inject noise into objective measurements. Calculate if our misspecification score (residual and max likelihood) is correlated to the strength of the induced misspecification. As a baseline I will use DWPI's raw inference loss.

References

[1] Lu, Mannion & Mason, "Inferring Preferences from Demonstrations in Multi-Objective RL", 2023.

[2] Bobu et al., "Learning under Misspecified Objective Spaces", CoRL 2018.

[3] Siththaranjan, Laidlaw & Hadfield-Menell, "Distributional Preference Learning", ICLR 2024.

[4] Bahlous-Boldi et al. "Pareto-Optimal Learning from Preferences with Hidden Context", NeurIPS 2025

[5] Settles and Burr, "Active Learning Literature Survey", 2009

Additional Note

Let me know if you have any ideas or feedback. I feel like there may be a better way to construct a scalar misspecification score and a better way to evaluate detection than correlation with some measurement of misspecification strength.

Additional Idea: Dynamic Hedging Against Misspecification

If we can reliably detect misspecification, I would love to dynamically hedge against assigning a misspecified policy. I think we could add weight to a MaxEnt objective or use distributional preference inference with CVaR-optimal policy assignment [3]. However, it would make this project too big in scope and depends on having good misspecification detection in the first place, so I'll just save the idea for later.